Week 7: Network Psychometrics

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1 Introduction

> **Learning objectives**
>
> By the end of this practical, you should be able to:
>
> 1. Explain the core concepts of network modelling, including nodes, edges, network structure, and common network types used in social and behavioural research.
> 2. Interpret key network metrics (e.g., degree, strength) and describe what they reveal about the structure and dynamics of a system.
> 3. Construct and visualise basic networks using R.
> 4. Apply introductory network-analytic techniques to real data, evaluate emerging patterns, and communicate findings using appropriate visual and statistical summaries.

In the lecture, we saw how network models represent psychological variables as nodes, and their conditional dependencies as edges. We also introduced regularisation to estimate sparse, interpretable networks, and bootstrapping to evaluate the stability of estimated edges.

Today, we will apply these ideas to two short questionnaires from a study of parents of children with special educational needs during the COVID-19 pandemic:

- **Part A:** Six items from the Kessler Psychological Distress Scale
- **Part B:** Six items from the Warwick–Edinburgh Mental Wellbeing Scale
- **Part C:** Combining the 12 items and figuring out the overall structure of Mental Health

2 Setting up

2.1 Packages

Install and load the following packages:

```
library(haven)      # import Stata data
library(tidyverse)  # data wrangling
library(psych)      # polychoric correlations
library(bootnet)    # network estimation + bootstrapping
library(qgraph)     # network visualisation
```

```
library(igraph)     # graph tools
library(NetworkComparisonTest) #network comparison test
```

3 Part A: Core Example

3.1 Introduction to the dataset

We begin with six distress items,these are taken from the Kessler Psychological Distress Scale

- nervous
- hopeless
- restless
- nothing_cheer
- everything_effort
- worthless

These are ordinal (1–5). Clear your workspace, set your working directory, and load the data - note that you'll need to use the `read_dta()` function to import this file.

```
# rm(list = ls())
# cat("\014")
#
# setwd("~/Library/CloudStorage/GoogleDrive-umar.toseeb@york.ac.uk/My Drive/Teaching 2025-26/Netw

data <- read_dta("./data/dataset.dta")
```

Next, extract the distress items from the dataset:

```
distress <- c("nervous", "hopeless", "restless",
              "nothing_cheer", "everything_effort", "worthless")

items <- data[distress]
nrow(items)
summary(items)
colMeans(items, na.rm = TRUE)
```

1. **How many observations are in the dataset?**
   Use nrow(items).
2. **How many missing values does each item have?**
   Use summary(items).
3. **Which item has the highest mean?**
   Use colMeans(items, na.rm = TRUE).

3.2 Preparing the data

Convert labelled variables to numeric to be sure:

```
items <- data.frame(lapply(items, function(x) as.numeric(as.character(x))))
str(items)
summary(items)
```

1. **Are all variables now numeric?**
2. **Have the descriptive statistics changed?**

3.3 Polychoric correlations

Compute polychoric correlations:

```
poly_corr <- polychoric(items)$rho
poly_corr
```

1. **Which pair shows the strongest correlation?**
2. **Which pair shows the weakest?**
3. **Do these patterns make psychological sense?**

3.4 Estimating a partial correlation network

```
net <- estimateNetwork(
  items,
  default   = "EBICglasso",
  corMethod = "cor_auto"
)
```

1. **Why is cor_auto appropriate?**
2. **What does the sparsity warning imply?**
3. **What is EBICglasso, and why is it used to estimate the network?**

3.5 Inspecting the partial correlation matrix

```
partial_cor_matrix <- net$graph
partial_cor_matrix
```

1. **Which edges are non-zero?**
2. **Which strong marginal correlations disappear?**
3. **What does this tell you?**

3.6 Plotting the network

```
qgraph(
  net$graph,
  labels          = colnames(items),
  layout          = "spring",
  edge.labels     = TRUE,
  edge.label.cex = 0.8
)
```

1. **Which symptoms are most strongly connected?**
2. **Are there clusters?**
3. **Does the visual structure match the matrix?**

3.7 Bootstrapping edge accuracy

```
boot_edges <- bootnet(
  net,
  nBoots = 100,
  type   = "nonparametric"
)

plot(boot_edges, "edge", plot = "interval")
```

1. **Which edges have narrow CIs?**
2. **Which are unstable?**
3. **How does this affect interpretation?**

3.8 Node centrality - Degree & Centrality

```
# Node degree = number of non-zero edges per node
degree <- apply(net$graph != 0, 1, sum)
sort(degree, decreasing = TRUE)

# Node strength centrality
cent <- centrality(net)
sort(cent$strength, decreasing = TRUE)

centralityPlot(net)
```

1. **Which symptom has highest degree?**

2. **Which has highest strength?**
3. **Do degree and strength tell a similar story?**
4. **How does this relate to edges?**
5. **Which symptoms might be leverage points to improve psychological distress?**

3.9 Stability of centrality

```r
# Case-dropping bootstrap for centrality stability
boot_cent <- bootnet(
  net,
  nBoots = 100,
  type   = "case"
)

# Plot centrality stability
plot(boot_cent, "strength")
```

```r
plot(boot_cent, "edge")
```

```r
corStability(boot_cent)
```

1. **Does node strength remain consistent?**
2. **Do strongest edges stay stable?**
3. **Are edge weights and strength stable enough?**
4. **Is CS $>$ .25?**
5. **What does this imply?**

3.10 Network Comparison Test: High vs Low Income

We now test whether the psychological distress network during COVID-10 differs between those from low income households and those from low income households:

- **High-income** (low_income = 0)
- **Low-income** (low_income = 1)

The NCT compares:

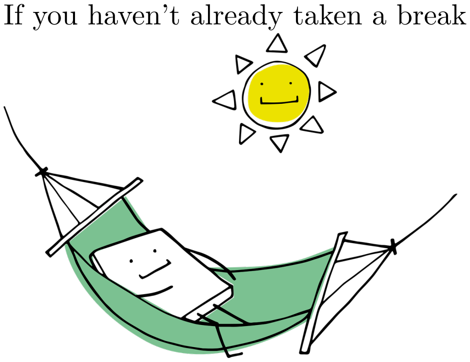
- **Global strength**
- **Network structure**
- **Edge differences**

```r
# Distress items - prepare the groups
distress <- c("nervous", "hopeless", "restless",
              "nothing_cheer", "everything_effort", "worthless")

# High-income group
items_high <- data[data$low_income == 0, distress]

# Low-income group
items_low  <- data[data$low_income == 1, distress]

#clean the data
items_high_clean <- na.omit(items_high)
items_low_clean  <- na.omit(items_low)

#run the NCT
nct <- NCT(
  data1 = items_high_clean,
  data2 = items_low_clean,
  test.edges = TRUE,
  test.centrality = FALSE,
  it = 1000
)

summary(nct)
```

1. **Is there a significant difference in global strength?**
2. **Is there a difference in overall network structure?**
3. **Are any edges significantly different?**
4. **What does this suggest?**

Take a break!

If you haven't already taken a break today, we suggest you do so before you start the next section.



4 Part B: Apply Your Knowledge

4.1 Introduction to the dataset

You now need to repeat the same workflow using six wellbeing items from the Warwick–Edinburgh Mental Wellbeing Scale:

- optimistic
- useful
- deal_problems
- think_clearly
- feeling_close
- make_up_mind

5 Part C: Challenge Yourself — Combined Network

In the earlier sections, you estimated separate networks for six distress items and six wellbeing items. In this challenge, you will combine all 12 items into a single dataset and estimate a joint network model. This allows you to explore how distress and wellbeing processes interrelate within the same system.

You will then use igraph to identify clusters (communities) in the combined network.

5.1 Prepare data

```r
# Vector of variable names
mental_health <- c("nervous", "hopeless", "restless",
          "nothing_cheer", "everything_effort", "worthless",
          "optimistic", "useful", "deal_problems", "think_clearly",
          "feeling_close", "make_up_mind")

# Subset the actual data from your dataset
items <- data[mental_health]

str(items)
summary(items)

# Ensure numeric type
items <- data.frame(lapply(items, function(x) as.numeric(as.character(x))))
```

5.2 Estimate the 12-item network and Plot it

```r
net <- estimateNetwork(items,
                       default = "EBICglasso",
                       corMethod = "cor_auto"
                       )

qgraph(
  net$graph,
  labels = colnames(items),
  layout = "spring",
  edge.labels = TRUE,        # <-- this adds the numbers
  edge.label.cex = 0.8       # <-- adjust text size (optional)
)
```

5.3 Convert to igraph and detect clusters using various different algorithms

```r
W <- net$graph # <-- convert to absolute values to get rid of negative values
W_abs <- abs(W)  # <-- convert to absolute values to get rid of negative values

g <- graph_from_adjacency_matrix(W_abs, mode = "undirected", weighted = TRUE, diag = FALSE)

wc <- cluster_walktrap(g, weights = E(g)$weight)
lv <- cluster_louvain(g, weights = E(g)$weight)
fg <- cluster_fast_greedy(g, weights = E(g)$weight)
im <- cluster_infomap(g, e.weights = E(g)$weight)
le <- cluster_leading_eigen(g, weights = E(g)$weight)
```

5.4 Inspect Clusters

```r
membership(wc)
membership(lv)
membership(fg)
membership(im)
membership(le)
```

5.5 Plot each solution

```r
plot(wc, g)
```

```r
plot(lv, g)
```

```r
plot(fg, g)
```

```r
plot(im, g)
```

5.6 Compare the number of clusters

```r
length(unique(membership(wc)))
length(unique(membership(lv)))
length(unique(membership(fg)))
length(unique(membership(im)))
length(unique(membership(le)))
```

1. **How many clusters does igraph detect?**
2. **Do distress and wellbeing items mix together or separate cleanly?**
3. **What does this tell you about the structure of mental health?**

6 End of practical

> **!** Feedback!
>
> These practical sessions are new this year. Please take a moment to rate the difficulty and length of these practical activities via this survey, and let us know of any issues you encountered or aspects you didn't understand (positive feedback is also welcome!). You will need to be logged in with your university account to respond, but your submission will be anonymous.
> For more general questions about the practical, please post them on the VLE Discussion Board.